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UAS Data Analytics

1. **NOTE:** *The dataset used in our UTS is different with the one we used in completing our project and this UAS. The previous dataset, Telco Churn Dataset, lacks the important feature for time series analysis since it was a snapshot dataset. After finding out about this while doing our UTS, we decided to use a transactional dataset, Rossmann Sales Dataset, and ended up doing the whole analytical process from the start. We are unaware that the first question in our UAS will ask us again about the result found in UTS. So, to address this issue, we decided for question number one, we will give two answers. One based on the result found from UTS using the old dataset; The other is based on the result found from completing our project with the new dataset. It is to be noted that the rest of the answers are based on the new analytical process using the new dataset. To Ko Rico, our lecturer, we apologized for this inconvenience and our lack of competence when initially selecting the dataset and for the decision to change it mid-semester.*

NO. 1 [BASED ON ANALYTICAL PROCESS DONE IN UTS]

a. Summarize the main data quality issues identified during the Mid Project (UTS)

- **Missing Values:** The Churn Reason column contained a very high volume of missing values, precisely 5,174 missing rows, making up roughly 73.46% of the entire dataset.
- **Incorrect Data Types and Structural Inconsistencies:** The column Total Charges was incorrectly recognized as an object data type rather than a continuous numeric value. This occurred because blank spaces (' ') were embedded within the column to represent certain accounts, which inherently obscured structural consistency and blocked arithmetic processing.

- **Irrelevant and Redundant Columns:** There were multiple geographic and tracking parameters (CustomerID, Count, Country, State, Lat Long, Zip Code, Latitude, Longitude, Churn Value) that offered redundant predictive signals or non-actionable baseline factors for subsequent exploratory data analysis (EDA).
- **Absence of Anomalies:** Initial checks confirmed that standard numeric columns (Tenure Months, Monthly Charges, Churn Score, CLTV) contained **0 rows with outliers** according to the Interquartile Range (IQR) bounds. Additionally, there were **0 duplicate records** overall.

b. Evaluate how effective your data wrangling process was in improving data quality

- It selectively dropped the massive sequence of geographic parameters and index variables, focusing cleanly on demographic and billing predictors.
- It systematically targeted the blank string structures blocking data conversion, converting implicit structural anomalies into cleanly mapped attributes.
- As a result of this wrangling sequence, the dataset was transformed from a raw tracker format into an optimization-ready model schema optimized for predictive EDA.

c. Support your evaluation using appropriate indicators (e.g., reduction in missing values, improved consistency, reduced outliers).

- **Data Type Realignment & Consistency Improvement:** The space characters (' ') inside Total Charges were dynamically converted to explicit numerical structures via `replace(' ', 0)` and finalized using `pd.to_numeric()`, shifting its data type from object to a usable float64 continuous type.
- **Dimensional Noise Reduction:** Dropping 5 categorical variables (CustomerID, Lat Long, Country, State, Churn Reason) and 5 numerical fields (Count, Zip Code, Latitude, Longitude, Churn Value) systematically trimmed the dataset to isolate high-value feature dimensions.
- **Missing Values & Outlier Stability Maintenance:** Missing rows under Churn Reason (73.46%) were structurally dropped as part of feature

pruning rather than being imputed with biased central tendencies. The remaining baseline statistics kept a clean count of **7,043 rows**, demonstrating a stable post cleanup dataset with no synthetic outlier creation or duplication distortions.

NO. 1 [BASED ON ANALYTICAL PROCESS DONE FOR OUR FINAL PROJECT]

a. Summarize the main data quality issues identified during the Mid Project (UTS)

- **Missing Values:** The dataset exhibited a substantial volume of missing data, particularly within the Promo2SinceWeek, Promo2SinceYear, and PromoInterval features, which each accounted for approximately 49.9% of the total observations. Additionally, CompetitionOpenSinceMonth and CompetitionOpenSinceYear contained missing values across 31.8% of the records, while a negligible volume of missing data was identified within the CompetitionDistance column. These missing values presented an immediate barrier to exploratory analysis, requiring systemic data cleaning and imputation.
- **Incorrect Data Types and Structural Inconsistencies:** Upon initial ingestion, the Date column was improperly parsed as an object/string data type rather than a functional datetime format. Furthermore, the StateHoliday column exhibited structural inconsistencies, characterized by a mix of numeric zeros (0) and categorical string labels, which required formatting standardization.
- **Irrelevant and Non-Operational Data (Noise):** The dataset contained an extensive sequence of records where stores were closed (Open = 0) and corresponding sales were zero. These observations introduced non-actionable baseline noise that offered no predictive signals for forecasting models or meaningful insights during exploratory data analysis (EDA).
- **Presence of Anomalies (Outliers):** Initial diagnostic checks utilizing the Interquartile Range (IQR) method identified outliers across standard

numeric features. Specifically, anomalous observations were detected within the Sales (2.6%), Customers (3.7%), and CompetitionDistance (9.7%) columns, establishing upper and lower bounds that flagged data points deviating significantly from the main distribution.

b. Evaluate how effective your data wrangling process was in improving data quality

- **Strategic Feature Pruning:** This pipeline proved highly effective, as it selectively dropped the redundant Open feature and systematically excluded records from closed store dates. By stripping away these non-informative observations, the workflow allowed subsequent modeling and analysis to focus exclusively on active sales predictors.
- **Systematic Structural Correction:** The pipeline systematically targeted string variables that impeded time-series modeling, transforming them into structured temporal features including Year, Month, WeekOfYear, and Quarter. Concurrently, missing data was handled by resolving structural gaps such as imputing missing promotional indicators with zeros or explicitly labeling them as 'none' ensuring a complete and consistent dataset for analysis.
- **Smart Imputation Over Deletion:** To prevent data loss from dropping rows, we applied targeted imputation across the dataset. We filled missing CompetitionDistance values with the median to minimize bias, while empty promotional features were cleanly mapped to structural constants like 0 or 'none'.
- **Optimization for Analytics:** Through this data wrangling sequence, the dataset was successfully transformed from a raw operational tracker into an optimization-ready model schema. This structured format establishes a clean foundation perfectly suited for rigorous exploratory data analysis (EDA) and robust regression modeling.

c. Support your evaluation using appropriate indicators (e.g., reduction in missing values, improved consistency, reduced outliers)

- **Data Type Realignment & Consistency Improvement:** The string structures within the Date column were dynamically transformed into

explicit temporal features. Furthermore, inconsistencies within the StateHoliday variable were systematically standardized by mapping numeric zeros (0) to a distinct 'no_holiday' label, thereby converting the underlying data types into fully functional, model-ready formats.

- **Dimensional Noise Reduction:** The process proved highly effective by removing records where stores were closed (Open = 0) or sales were flat (Sales = 0), and subsequently dropping the redundant Open column. This cleanup successfully trimmed the dataset to isolate high signal features and eliminate operational noise.
- **Missing Values & Outlier Stability Maintenance:** Instead of dropping records, missing values in CompetitionDistance were filled using the median to maintain an unbiased central tendency. The post cleanup baseline statistics show a stable, well distributed dataset that is fully ready for predictive modeling without the risk of synthetic outliers or duplication distortions.

2. Advance Pattern, Trend, and Relationship Analysis

a. Examining The Variables Deeper

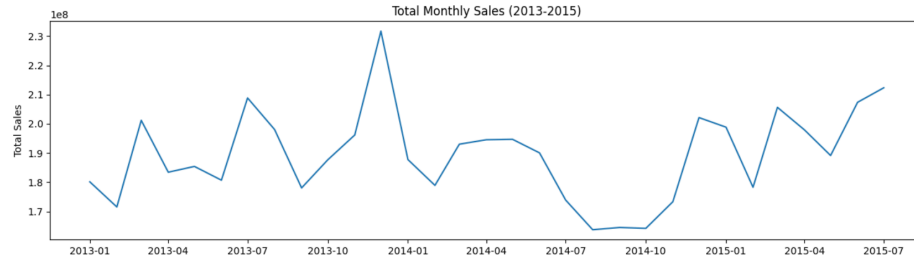
- **Segmented Analysis: Promo Effect Across Store Types**

A segmented analysis was conducted to evaluate how promotional impact varies across different store types. Initial A/B testing confirmed that promotions significantly drive sales ($U = 45,073,973,080$, $p < 2.2e-16$), with the treatment group (Group B) achieving a 38.8% higher mean sales lift than the control group (Group A). However, while promotions consistently boost sales across all categories, a uniform strategy may not yield optimal results. To maximize performance, the Rossmann team should develop tailored promotional plans customized to each specific store type.

- **Time Based Trends and Seasonal Patterns**

Time-series analysis from 2013 to 2015 reveals a consistent annual seasonal pattern, where sales peak during November and December driven

primarily by holiday shopping. At a granular level, daily sales exhibit a weekly trend: performance peaks on Mondays and gradually declines toward the weekend.



Notably, the data highlights an anomaly between July and October 2014, where sales declined significantly diverging from the established 2013 baseline. Investigation confirmed this drop did not correlate with changes in active store counts or promotional volume, pointing to external factors. However, a sharp recovery occurred in November 2014, coinciding with a strategic expansion of promotional activity from approximately 39% to 61% of stores. This demonstrates that promotions function effectively as a recovery mechanism rather than just a routine sales driver.

- **Subgroup Comparison: Sales Across Store Types**

A subgroup comparison across store types reveals a striking disparity in performance. StoreType B generates the highest average daily sales (10,233), substantially outperforming all other store types, which range between 6,822 and 6,933. This finding is particularly notable given that StoreType B is the least represented group in the dataset, indicating its high performance is independent of sample frequency. A Kruskal-Wallis test confirms that these variations are statistically significant ($H = 8,227.55$, $p < 2.2e-16$), validating that the observed differences do not stem from random variation.

b. The Most Influential Variables

To identify the most influential variables, we utilized two complementary approaches: Mutual Information (MI) Score and Random Forest Feature

Importance. Both methodologies converged on the same conclusion: the two features engineered during the second modeling iteration to enhance accuracy StoreMonthAvg and StoreAvgSales are the most informative predictors. Specifically, StoreMonthAvg alone accounts for a Random Forest importance score of approximately 0.7. This demonstrates that historical sales patterns aggregated at the store-month level serve as the strongest predictors for forecasting future sales.

c. Why These Relationships are Meaningful

The analytical relationships identified directly address the core business objective: understanding the underlying drivers of sales variance across Rossmann stores. The primacy of StoreMonthAvg as the most influential feature reveals that each location possesses deeply ingrained historical sales rhythms. This indicates that future sales performance is primarily shaped by a store's baseline trajectory rather than external variables alone, rendering store-level historical context indispensable for accurate forecasting. Consequently, any predictive or planning model that omits this historical baseline will likely yield unreliable projections. Furthermore, the significance of Promo as a supporting feature validated via both A/B testing and feature importance metrics confirms that promotional activity is a consistently reliable sales driver rather than a mere correlation. Finally, the statistically significant performance gap between StoreType B and other store types, verified by the Kruskal-Wallis test, underscores structural or operational variations that meaningfully alter sales outcomes across different store formats.

3. Evaluation of Analytical Methods and Metrics

a. Analytical Techniques and Models Used

- Statistical Analytical Techniques

For this phase, we deployed two non-parametric statistical methods. An A/B test was executed using the Mann-Whitney U test to evaluate sales performance differences between promotional and non-promotional conditions. Concurrently, the Kruskal-Wallis test was utilized to conduct multi-group comparisons across different store types. Both methods were

specifically selected for their robustness against non-normally distributed data, addressing the distinct right-skewed distribution observed in the sales dataset.

- **Predictive Modeling**

In the modeling phase, we selected the Random Forest Regressor as our primary predictive model. This choice was driven by the algorithm's inherent robustness to outliers and its capacity to effectively capture complex, non-linear relationships between the input features and the target sales variable.

- **Supporting Techniques**

To optimize model training, we implemented several preprocessing and optimization steps. Label Encoding was executed to convert categorical variables into numerical representations, while a Standard Scaler was utilized to normalize numerical features. For feature selection, we applied Mutual Information (MI) Scoring, retaining only features that yielded a score above zero. Finally, hyperparameter tuning was conducted using RandomizedSearchCV on a 10% sample of the training dataset to effectively manage computational constraints.

b. Appropriate Performance or Evaluation Metrics

- **Statistical Testing Metrics**

For both the hypothesis and A/B testing phases, the p-value served as the primary evaluation metric against a significance threshold (α) of 0.05. A p-value below this alpha level indicates that the observed differences between groups are statistically significant and unlikely to be the result of random variation. Additionally, the U statistic (from the Mann-Whitney test) and the H statistic (from the Kruskal-Wallis test) are reported to quantify the magnitude and direction of the test results.

- **Predictive Model Metrics**

To evaluate the predictive model, we utilized three distinct metrics: MAE, RMSE, and R^2 . Mean Absolute Error (MAE) measures the average absolute deviation between predicted and actual sales,

providing an intuitive sense of error in the same unit as the target variable. Root Mean Squared Error (RMSE) serves a similar purpose but applies a higher penalty to larger errors, acting as an indicator of model stability. Finally, the Coefficient of Determination (R^2) measures the proportion of variance in sales explained by the model; our score of 0.8162 demonstrates that the model accounts for approximately 82% of the total variance.

Notably, this performance was achieved through an iterative, two-cycle analytical process. Prior to the second iteration which introduced key engineered features like StoreMonthAvg the initial R^2 score was only 0.4743. These metrics not only quantified current model performance but also guided the overall refinement of our analytical workflow to produce a substantially more accurate predictive model.

c. Limitations

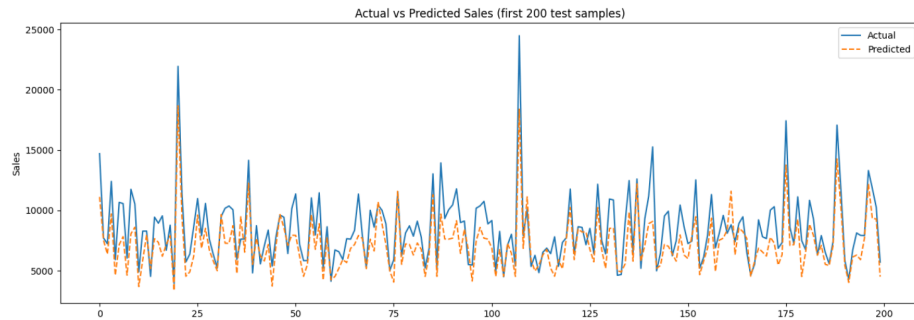
- Statistical Testing Limitations

While both the A/B test and the Kruskal-Wallis test yielded statistically significant results, it is critical to note that the exceptionally small p-values ($p < 2.2e-16$) are heavily driven by the large dataset size (approximately 844K observations). At this scale, even negligible differences between groups can trigger statistical significance. Consequently, practical significance such as the observed 38.8% sales lift in the A/B test must be evaluated alongside statistical metrics to draw meaningful conclusions. Our analysis balances both dimensions: statistical significance confirms that the variations are not due to random chance, while practical significance ensures the insights hold tangible business value.

- Predictive Model Limitations

In our second modeling iteration, the model achieved a high R^2 score of 0.8162. However, as illustrated in the visualization, it consistently

underestimates extreme sales spikes a limitation that could compromise forecasting accuracy during peak periods like holidays. This behavior stems from an inherent constraint of Random Forest models when trained on datasets where extreme values are underrepresented relative to the broader distribution. Furthermore, because hyperparameter tuning was restricted to a 10% sample of the training data due to computational constraints, the selected parameters may not fully generalize to the entire dataset. This sample limitation partly explains why the tuning process failed to yield a substantial performance improvement over the baseline model.



4. Exploratory Data Analysis (EDA)

- First Recommendation

a. Actionable Recommendation

Our analytical findings demonstrate that promotions generate an average sales lift of 38.8% compared to non promotional periods. Furthermore, historical data confirms that promotions serve as an effective recovery mechanism; in November 2014, a sharp increase in promotional coverage (from 39% to 61% of stores) successfully reversed a prolonged sales downturn triggered by external factors between July and October. Based on these insights, we recommend that Rossmann establish an early warning monitoring system to detect abnormal sales dips. This will allow the team to deploy targeted promotions as a reactive, structured intervention to mitigate revenue declines, shifting away from rigid, fixed calendar scheduling.

b. The Effect to Decision Making

This framework provides the Rossmann management team with a data-backed trigger mechanism. By utilizing historical StoreMonthAvg data to establish an expected baseline, the system can automatically flag instances where actual sales fall below performance thresholds. This early-warning alert grants the team crucial lead time to isolate and investigate the root cause whether internal or external. Consequently, promotional deployment transforms into a targeted mitigation strategy executed while operational corrections occur concurrently. This approach successfully shifts corporate decision making from intuition based to evidence based, while preserving vital human judgment in the final execution call.

c. Potential Risks

1. When a sales downturn is driven by external market forces as observed during the 2014 anomaly relying exclusively on promotional strategies may prove insufficient to achieve a complete revenue recovery.
2. Implementing frequent, reactive promotions without stringent discount controls risks eroding profit margins over the long term, potentially compromising sustainable revenue growth.
3. Repeated exposure to promotional pricing risks shifting consumer reference points; over time, customers may perceive discounted rates as the baseline price. This psychological shift diminishes the long-term elasticity and efficacy of promotional campaigns while structurally depressing potential top line revenue.
4. Excessive alert generation within the monitoring system risks inducing notification fatigue; over time, the operational team may begin to underrespond to or dismiss warnings entirely. This behavioral desensitization structurally undermines the integrity and long-term efficacy of the early-warning mechanism itself.

- Second Recommendation

a. Actionable Recommendation

Analysis indicates that StoreType B consistently records the highest average daily sales at approximately 10,152, significantly outperforming other store categories which cluster between 6,800 and 6,900. This variation is highly notable given that StoreType B is the least represented category in the dataset, demonstrating that its superior performance cannot be attributed to scale or frequency alone. Consequently, we recommend that the Rossmann business team conduct a targeted operational investigation into StoreType B locations. Granular factors such as microlocation characteristics, store footprints, product assortment strategies, and staffing models should be evaluated to isolate the primary drivers of this outstanding performance, enabling the potential replication of these success factors across other compatible store formats.

b. The Effect to Decision Making

This recommendation provides a clear guidance for the Rossmann strategic and operations teams to start digging further into the stores internal benchmark. The data suggest that solutions in achieving better sales numbers can be found within the stores network. Understanding what makes StoreType B different gives leadership a concrete, evidence based foundation to inform decisions around store layout, assortment planning, and expansion strategy.

c. Potential Risks

1. The superior performance of StoreType B may be intrinsically tied to unique, non-replicable variables. If these underlying drivers are structurally distinct, exporting its operational framework to different store formats may fail to yield comparable performance gains.
2. Restructuring other store formats to mirror the characteristics of StoreType B may demand substantial capital expenditure while offering highly uncertain returns on investment.

3. Because StoreType B is the least represented category, it inherently suffers from a limited sample size. Consequently, strategic conclusions drawn from a restricted number of observations may lack the robust generalizability required for network wide implementation.